Sample Project Documentation

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Project Description:-

Human Development Index (HDI) Prediction Using Machine Learning

Project Description

The **Human Development Index (HDI) Prediction Using Machine Learning** project is designed to predict the Human Development Index category of a country using socioeconomic indicators. The application uses machine learning algorithms to analyze four important development factors:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) Per Capita

The system is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**. A trained machine learning model predicts whether a country belongs to the **Low, Medium, High, or Very High Human Development** category.

The web application provides an easy-to-use interface where users can enter the required values and instantly receive the predicted HDI category.

Project Scenarios

Scenario 1

A government officer wants to estimate the HDI level of a developing country after collecting the latest education, health, and income statistics. The system predicts the country's HDI category instantly.

Scenario 2

A university student studies the impact of education on human development. After entering educational indicators into the application, the predicted HDI category is displayed for research purposes.

Scenario 3

An NGO working on rural development compares different regions using development indicators. The application predicts HDI categories, helping identify areas that require additional support.

Team Members

Name	Role	Responsibility
G Naveen	Team Leader	Project Planning, GitHub Repository Management, Project Integration
Chandana Busireddy	Team Member	Dataset Collection, Data Cleaning, Documentation
Gandham Joshna	Team Member	Data Visualization, Testing, Report Preparation
Nandineni Gowthami	Team Member	Frontend Development (HTML, CSS, JavaScript), Flask Templates
Shaik Talha	Team Member	Machine Learning Model Development, Flask Backend, GitHub Integration, Model Testing

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Technologies Used

- Python
- Flask
- HTML5
- CSS3
- JavaScript
- Pandas

- NumPy
- Scikit-learn
- Joblib
- Visual Studio Code
- Git & GitHub

Technical Architecture

```
User
  │
  ▼
Web Browser
  │
  ▼
Flask Application
  │
  ├──────────────▶ Input Validation
  │
  ▼
Machine Learning Model (.pkl)
  │
  ▼
Prediction Engine
  │
  ▼
HDI Category
  │
  ▼
Display Result
```

Milestone 1

Requirement Analysis & Dataset Collection

Activities

- Study the Human Development Index.
- Collect HDI dataset.
- Understand project objectives.
- Select required features.
- Analyze dataset quality.
- Install required software.

Milestone 2

Data Preprocessing

Activities

- Handle missing values.
- Remove duplicate records.
- Encode categorical values.
- Normalize numerical features.
- Split dataset into training and testing datasets.

Milestone 3

Machine Learning Model Development

Activities

- Import Scikit-learn libraries.
- Train the model.
- Compare different algorithms.
- Select the best-performing model.
- Save the trained model using Joblib.

Milestone 4

Flask Web Application Development

Activities

- Design HTML interface.
- Create Flask routes.
- Load trained model.
- Accept user input.
- Predict HDI category.
- Display prediction result.

Milestone 5

Testing & Deployment

Activities

- Test the application locally.
- Verify prediction accuracy.
- Upload source code to GitHub.
- Deploy the application.
- Final project demonstration.

Home Page

The home page contains:

- Project title
- Introduction
- Input form
- Predict button
- Navigation menu

Prediction Page

Users enter:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI Per Capita

After clicking **Predict**, the application processes the information using the trained machine learning model.

Results Page

The results page displays:

- Input values
- Predicted HDI category
- Prediction confidence (optional)
- Success message
- Prediction summary

Conclusion

The **Human Development Index Prediction Using Machine Learning** project demonstrates how artificial intelligence and machine learning can assist in analyzing a country's development level using health, education, and economic indicators. The system provides quick and accurate predictions through a user-friendly Flask web application. This project can be enhanced in the future by integrating real-time datasets, cloud deployment, advanced machine learning models, and interactive dashboards for better visualization and decision-making.